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PrivAR: Client-Side Privacy Framework for Real-Time Location-Based Augmented Reality

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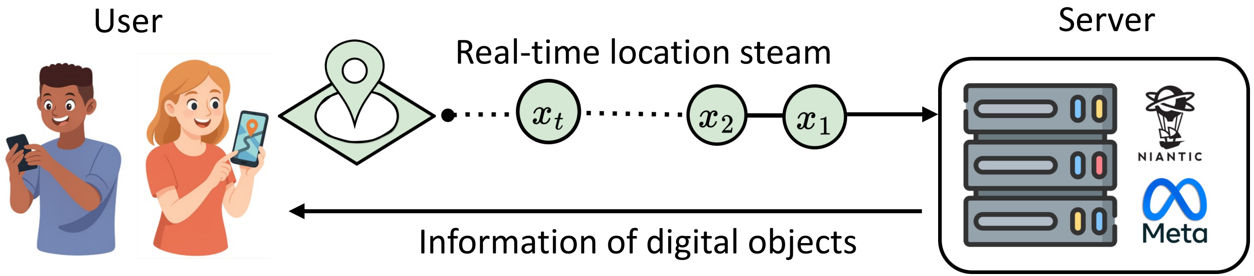
Location-Based Augmented Reality (LB-AR)

Augmented Reality (AR) overlays digital content onto physical word. Location-based AR (LB-AR) uses real-time location to decide where to place the digital content.



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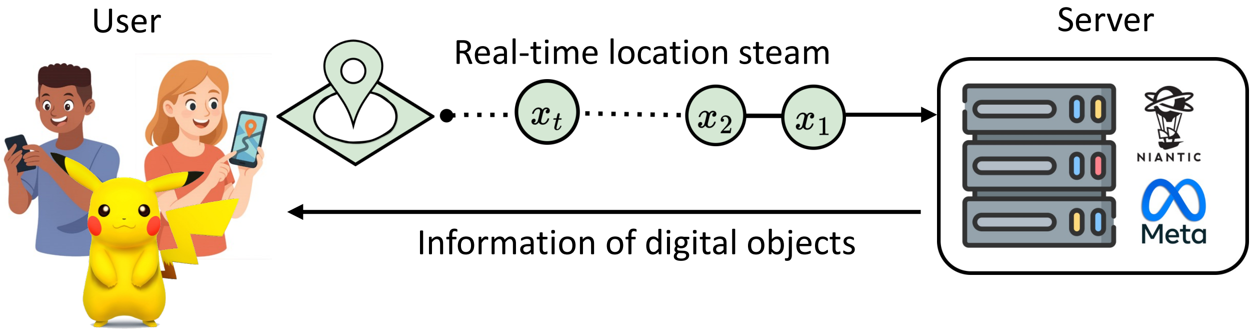

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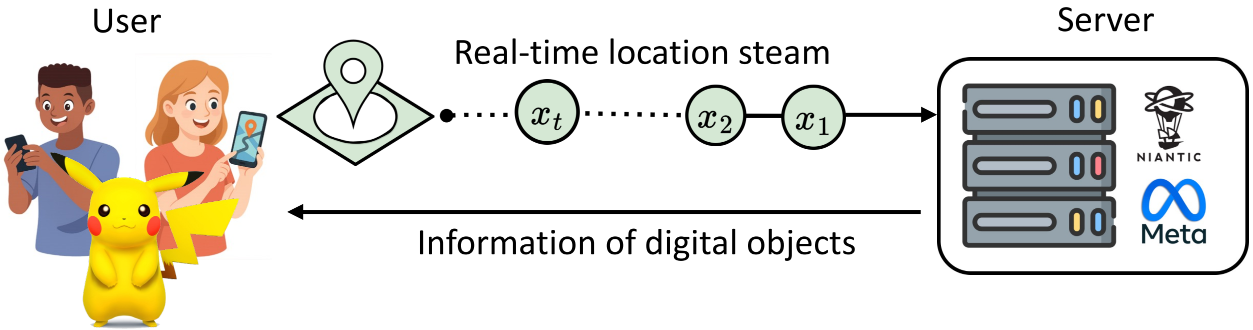

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  - This continuous high-frequency location requests makes LB-AR uniquely privacy-invasive

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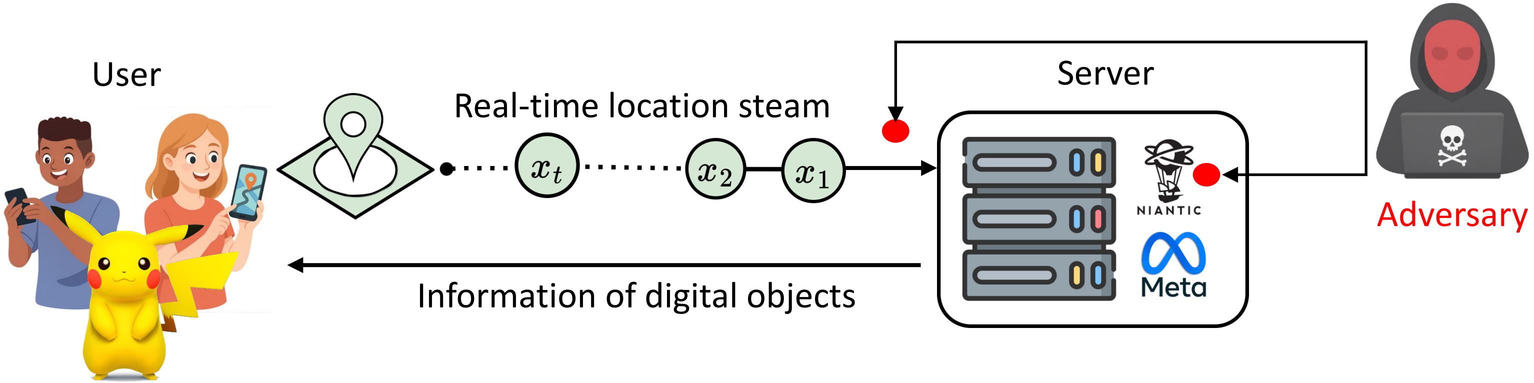

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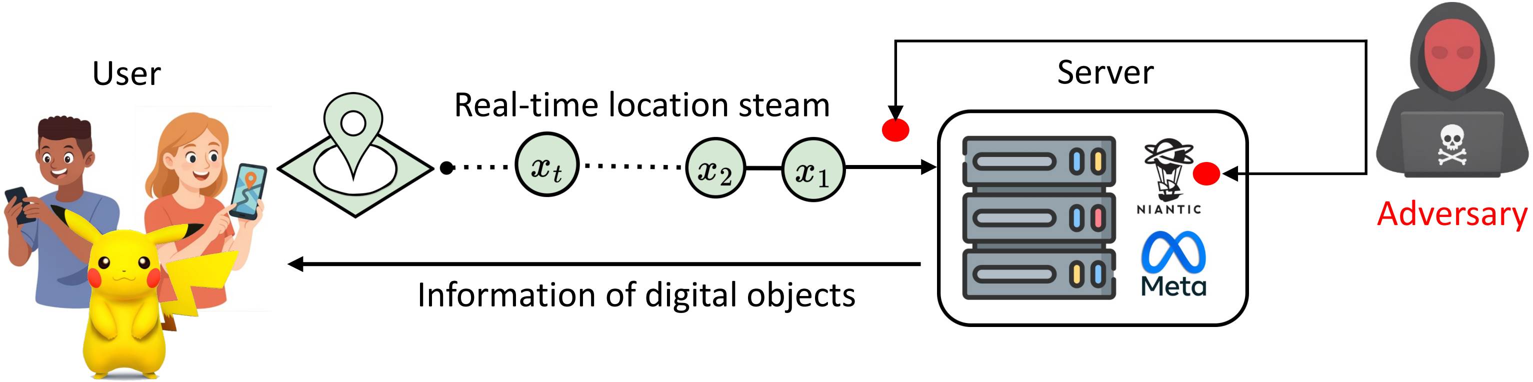

- Pokemon Go requests upto 13 location updates per minute from users for seamless gameplay
  - This continuous high-frequency location requests makes LB-AR uniquely privacy-invasive
- Adversaries use mobility data to infer even about user's daily routines, social circles, etc.
- Therefore, It's important to protect these high-frequency mobility data from adversaries

Privacy Mechanisms for LB-AR

An effective privacy mechanism for LB-AR must meet three requirements.

Strong Privacy	High Quality of Service (QoS)	Low Latency
• Per-location guarantee • Trace-level privacy	• Maintain immersive user experience	• Incurs milliseconds overhead only

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  - Spatial Cloaking does not provide strong privacy guarantee, Local Differential Privacy excessively harms QoS, and Cryptographic protocols incurs heavy computation cost

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- The Planar Laplace Mechanism (PLM), a <u>Geo- indistinguishability (GeoInd)</u> based privacy mechanism, is a reasonable choice for protecting LB-AR systems
  - Low-computation overhead and simplistic implementation

Planar Laplace Mechanism (PLM)

In PLM, instead of reporting true location x, the user reports a noisy location z as:

Euclidean distance between x and z

$$D_{\epsilon}(x)(z) = \frac{\epsilon^2}{2\pi} e^{-\epsilon d(x,z)} \quad \text{-------------------- (Eq. 1)}$$

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- PLM satisfies ϵ-GeoInd, providing formal privacy guarantee [1]
  - GeoInd ensures nearby locations should generate statistically similar perturbed locations
- PLM has low-latency, ideal for LB-AR that rely on high-frequency location updates

[1] M. E. Andr´es, N. E. Bordenabe, K. Chatzikokolakis, and C. Palamidessi, Geo-Indistinguishability: Differential Privacy for Location-Based Systems

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- However, PLM has two limitations, contributing to less QoS and poor Trace-level Privacy

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Limitations of PLM

For efficient on-device sampling, PLM is transformed to its the polar form. Let $z - x = r$, where r is the induced radius, and θ is the angle. Transforming Eq. 1 gives:

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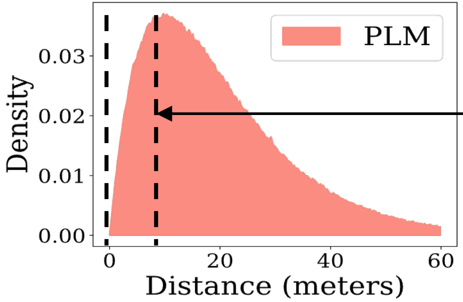

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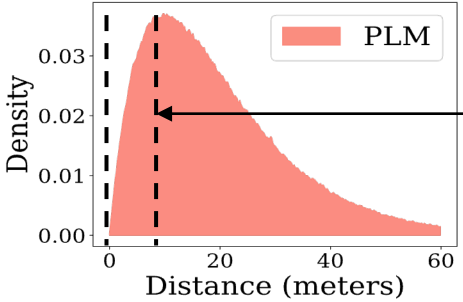

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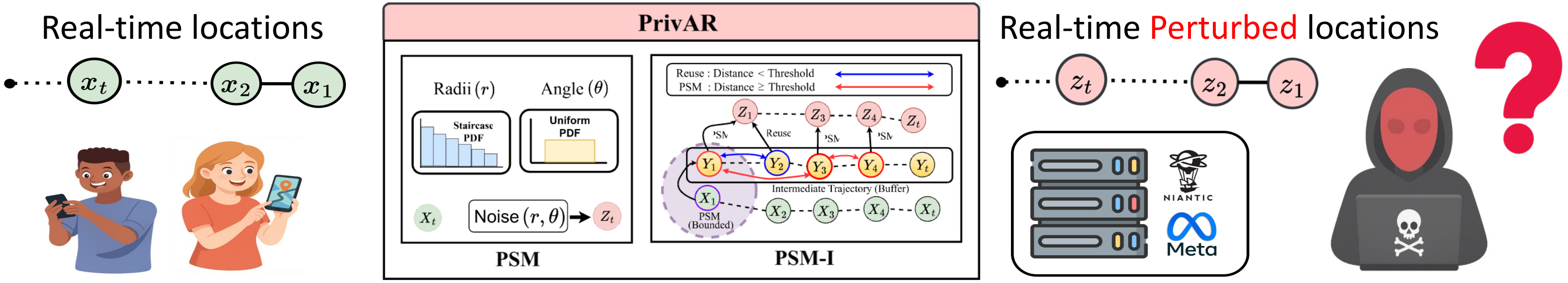
2. For trajectory $x = \{x_1, \ldots, x_T\}$, independently applying PLM quickly exhausts privacy budge Therefore, trave-level guarantee becomes weak.

PrivAR: Real-time Privacy Framework for LB-AR

We propose PrivAR, the first plug-and-play privacy framework for LB-AR apps that protects user's sensitive mobility data by perturbing location updates in real-time



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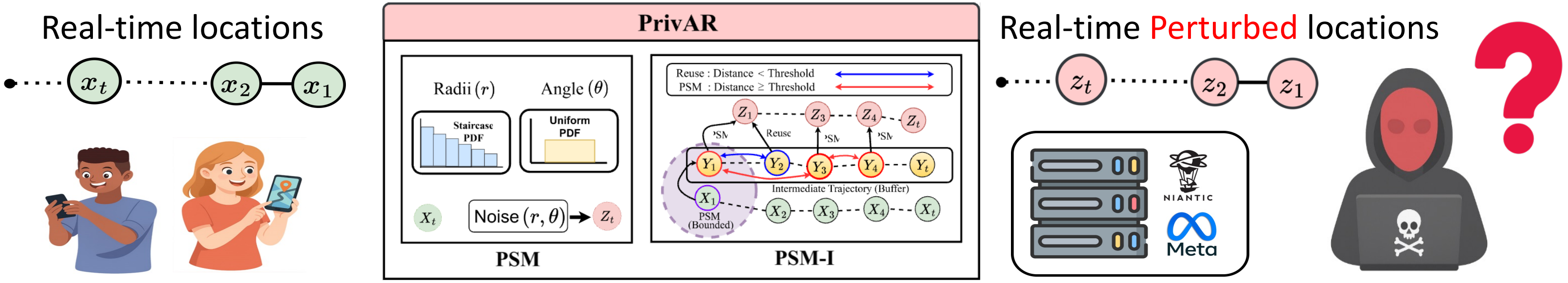

1. We propose, Planar Staircase Mechanism (PSM), which provides strong per-location privacy, high QoS, and low computation overhead.

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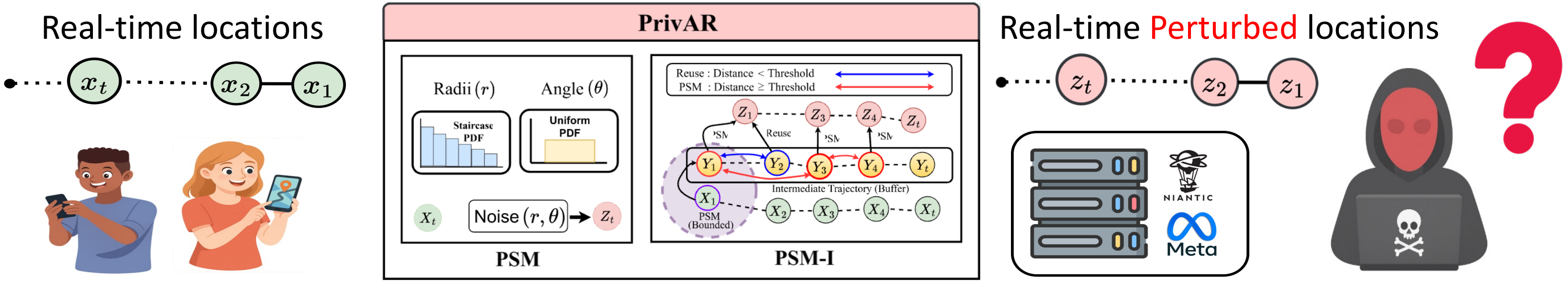


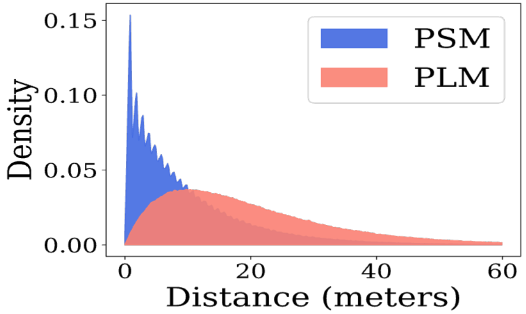
1. We propose, Planar Staircase Mechanism (PSM), which provides strong per-location privacy, high QoS, and low computation overhead.

2. We propose, Planar Staircase Mechanism with Intermediate (PSM-I), an extension to PSM, which provides strong trace-level protection.

Planar Staircase Mechanism (PSM)

A staircase-shaped distribution that favors nearby noisy locations.

- Higher mass near the true location improves QoS



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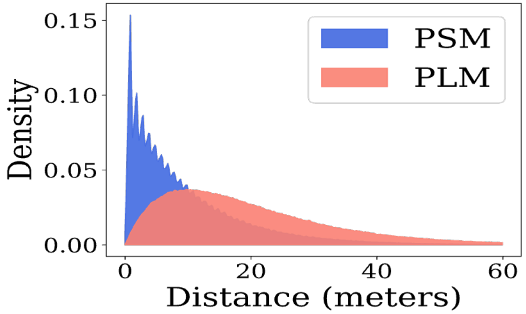
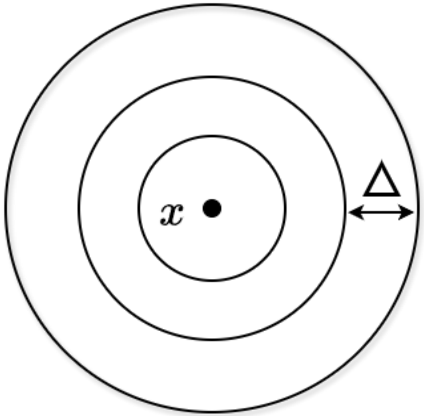


Construction of PSM:

- Divide the plane around x into circular rings of equal width Δ



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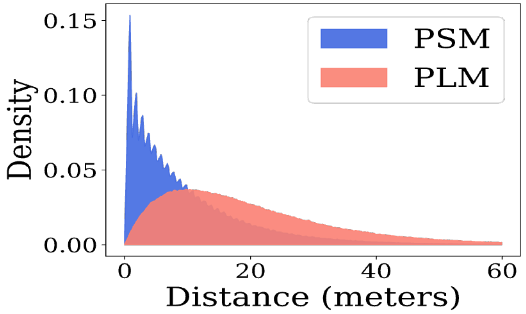
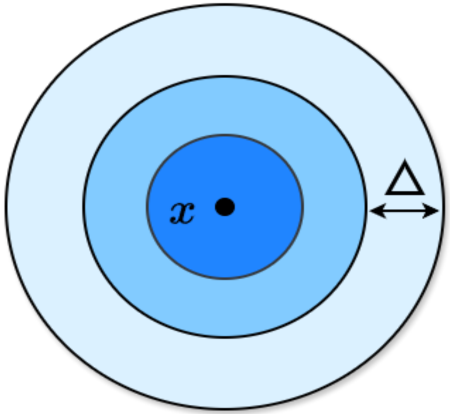


Construction of PSM:

- Divide the plane around x into circular rings of equal width Δ
- Choose a ring (I) using a geometric distribution
  - Nearby rings are more likely

$$\Pr[I = i] = (1-q)q^{i-1}, \qquad q = e^{-\epsilon\Delta}$$



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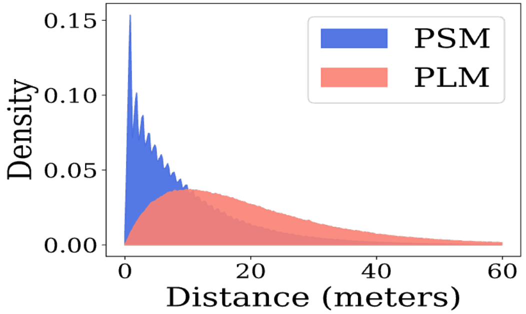

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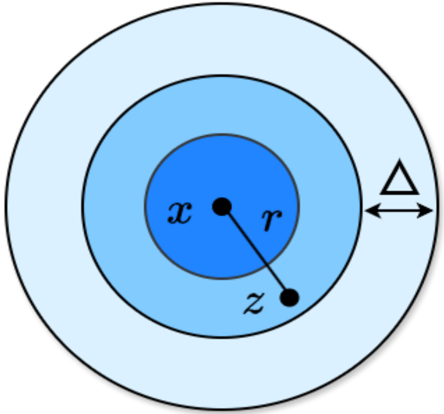
- Sample a point inside the chosen ring. Choose radius r and θ, report:

$$z = x + r(\cos(\theta), \sin(\theta))$$



Planar Staircase Mechanism (PSM)

- High QoS: Table confirms empirical displacement error under PSM is half that of PLM
  - Perturbed locations are more likely to be closer to true location, ensuring higher QoS.

Mechanism	Mean Distance (m)	Max Distance (m)
PLM	20.001	155.524
PSM	**10.044**	**141.719**

- Low Latency: Table confirms PSM and PLM have comparable latency
  - 0.05- 0.56 seconds, depending on device

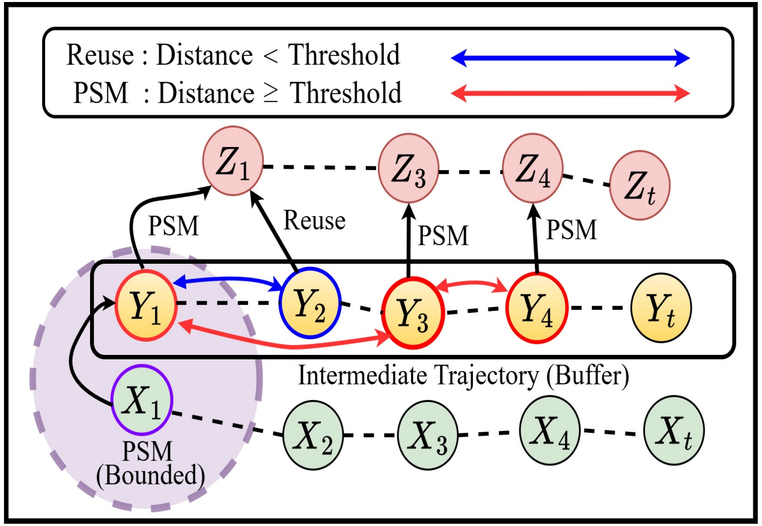
Device	**PLM**	**PSM**
Galaxy A04 (Low-end)	0.556	0.551
Pixel 6a (Mid-range)	0.114	0.112
Galaxy S22 Ultra (High-end)	0.046	0.048

- Formal Privacy Guarantee: PSM satisfies GeoInd
  - Please see Theorem 1 in the main paper

Planar Staircase Mechanism with Intermediate (PSM-I)

PSM-I ensures strong trace-level privacy using intermediate privacy buffer and selective refresh policy.

- Maintains intermediate buffer trajectory $\{y_t\}$
  - This reduces direct linkage between the released stream and the true stream. The adversary sees this buffer trajectory only.
- Perturb y_t with PSM if $distance \geq threshold$
  - multiple consecutive intermediate locations corresponds to the same released point, weakening temporal linkability.



Evaluation Setup

- We evaluate PrivAR using two public dataset: Geolife and T-Drive
  - Geolife: 23,061 km, with a median step of 8m and 2s intervals
  - T-drive: 809K points across 4,343 km, with a 49m median step and 181s intervals

- We developed a Pokemon-go inspired Android-app to further evaluate the practicality of PrivAR, using our collected dataset, GeoTrace
  - Five participants (3 male, 2 female), 127 km of trajectory data

- Three mobile devices (low-end to high end) to test the real-time practicality
  - Galaxy A04 (Low-end) , Pixel 6a (Mid-range), Galaxy S22 Ultra (High-end)

Evaluation Metrics

- We measure Quality of Service (QoS) using Mean Normalized Error (MNE)
  - Measures the average distance between true and perturbed locations
    - Lower is better

- We measure Privacy using Bayes Risk [2]
  - Measures the attacker's success of inferring the true location from perturbed location
    - Higher is better

- We measure Latency using the average time required to perturb a single location
    - lower is better

[2] F-bleau: fastblack-box leakage estimation," in 2019 IEEE Symposium on Security and Privacy (SP)

Evaluation : Public Datasets

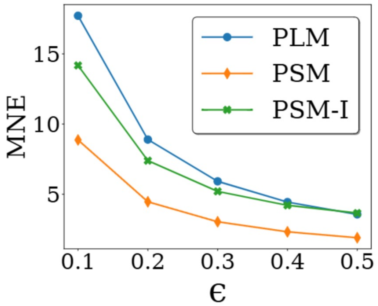
- QoS: PSM and PSM-I outperforms PLM, on evaluated privacy budgets

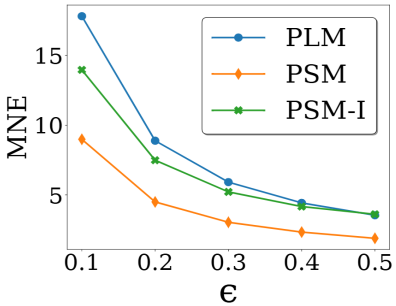
- Privacy: PSM-I outperforms PSM and PLM
  - PSM and PLM offer comparable levels of privacy protection

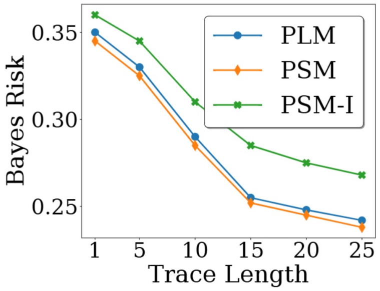
- Latency: PSM and PLM have comparable time
  - PSM-I slightly higher due to the intermediate step

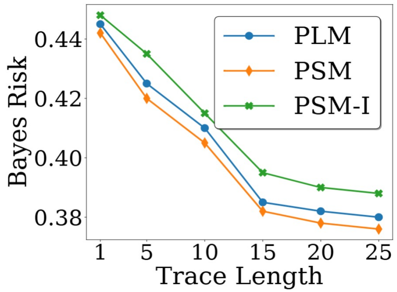
Geolife

T-Drive









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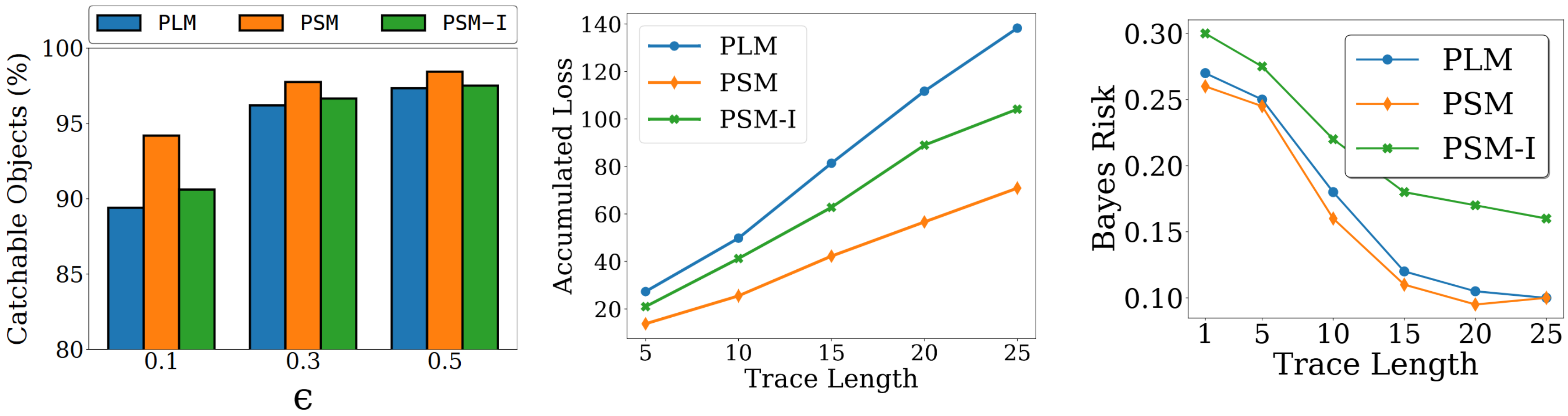
Pokemon-go Inspired Android App

We integrated PrivAR to a prototype app that let user's collect digital object in real-time.

- Allows users to select their privacy mechanism and their privacy level

- Allows users to see in real-time ratio of the digital object with/without privacy mechanism

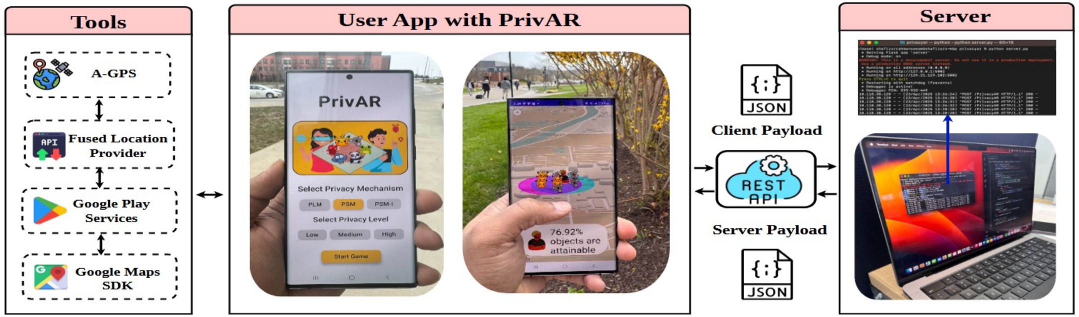
Evaluation: GeoTrace (Collected) Dataset



- Catchable Objects shows of virtual objects still reachable after perturbation
  - PSM and PSM-I better compared to PLM
- Accumulated loss shows of total number of objects missed over a session
  - PSM and PSM-I better compared to PLM
- In addition to improving the QoS, PSM-I provides better trace-level privacy
  - Bayes risk is higher PSM-I

Evaluation: End-to-End

Operation	Duration (ms)
Device acquires location fix	0.03
Apply PrivAR (PSM/PSM-I)	**0.06**
Serialize JSON payload	1.32
Server-side object generation	0.28
Network round-trip (request + response)	30.52
Parse JSON response and render object	1.70
Total End-to-End Latency	33.91



- End-to-end latency (33.91 ms) is dominated by network roundtrip (30.52 ms)
  - Privacy mechanism adds negligible time (0.06 ms, on Galaxy S22 Ultra)

Conclusion

- We propose PrivAR, the first work to systematically study client-side privacy protection for real-time LB-AR systems.

- Under PrivAR framework, we propose two privacy mechanism: PSM and PSM-I.
  - PSM reshaped the displacement distribution that improves QoS
  - PSM-I uses intermediate buffer for strong trace-level privacy

- We provide theoretical analysis, extensive evaluation on public dataset and our collected GeoTrace dataset, and validate PrivAR in a prototype mobile app.
  - Results show that PrivAR improves improves QoS, provides strong privacy guarantee, and only requires low computation cost, making it ideal for LB-AR settings

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Thank You!

Please email the authors if you have any question or suggestions.

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